A Cute Water Heater

Yes, there are cute water heaters. ECO-CUTE is a Japanese trademark and nickname for heat pump water heaters that use CO_2 as a refrigerant. (Cute is the pronunciation of “hot water supply” in Japanese. Whether the technology is cute or not is in the eye of the beholder.) Water heating is one of the biggest single energy consumers in Japanese households. Maybe that’s because of the climate, or maybe it’s because we Japanese love to take baths. It takes 50 gallons of hot water to fill a typical Japanese bathtub, and the typical single-family household uses around 100 gallons of hot water a day. That helps to explain why high-efficiency heat pump water heaters are so popular in Japan.

The ECO-CUTE water heater comes in two standard models. The 370 l tank, which holds approximately 100 gallons and is suitable for a family of three to five people, has a heating capacity of 4.5 kW. The 460 l tank, which holds approximately 120 gallons and is suitable for a family of four to seven people, has a heating capacity of 6 kW.

The heat pump unit and the tank unit are connected by water pipes and are installed outside the house. Both the heating capacity and the tank size of the ECO-CUTE are greater than those of U.S. heat pump water heaters, and the water temperature in the tank is higher. This is because we use time-of-use cheap electricity in the middle of the night to heat up and store most of the hot water consumed during the day (see Table).

Benefits of CO_2

CO_2 heat pump water heaters have several advantages over conventional water heaters. First, they are highly efficient—more than three times as efficient as conventional models, and just as efficient as typical heat pump water heaters. And CO_2 is more environmentally friendly than the hydrofluorocarbon (HFC) refrigerants that are used in conventional heat pump water heaters. This is because CO_2 does not deplete the ozone layer and contributes very little to global warming. Furthermore, CO_2 heat pump water heaters can heat water up to 194°F even if the ambient temperature is -4°F. This is because of the CO_2 heat pump cycle’s supercritical state and a small high-low pressure ratio, as compared to the R410A heat pump cycle, for example. HFC heat pump water heaters cannot function well at these low temperatures. Also, the higher pressure of the CO_2 heat pump cycle means a higher vapor density of the refrigerants in the evaporator. This allows for compact evaporators, and a smaller flow volume (and lower pressure drop loss) to get the same heating capacity.

ECO-CUTE heat pump water heaters have informational remote controllers that allow you to change the delivered water temperature digitally; fill a bathtub with hot water; check the remaining hot water in the tank; and know how much hot water you’ve consumed. Also the ECO-CUTE will learn how much hot water you consume per day and then automatically heat and store that amount of hot water. —And if you pay an extra thousand dollars, (on top of the $6,000–$9,500 basic price) you can get hydronic floor heating or a mist sauna function as well.

Table. Time of Use Rates, Tokyo Electric Power Company

TIME OF USE	RATE
11pm — 7 am	$ 0.07/kWh
7 am — 10 am	$ 0.21/kWh
10 am — 5 pm	$ 0.26 /kWh $ 0.31/kWh (Summer)
5 pm — 11 pm	$ 0.21/kWh

Japanese bathtub

The Midnight Hours

Electric utilities in Japan offer nighttime time-of-use plans with electricity rates one-third the regular, daytime rates. Since the ECO-CUTE is intended to operate mainly during the night, using this cheap electricity, operating cost can be as low as $10–$20 per month. This is about 20% of the operating cost of a gas water heater. And there is also a $420 government subsidy for CO_2 heat pump water heaters in Japan.

The shipment of ECO-CUTE water heaters has boomed since their debut in 2001. More than one million units have been installed so far. Although gas tankless water heaters still dominate the Japanese market, heat pump water heaters made up nearly 10% of all water heaters shipped in 2007.

—Takehiro Maruyama

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